Robot Dog Design

1. Body Structure (Frame)

A rectangular frame made of **aluminum** or **reinforced plastic**.
The following components are placed in the center of the frame:
Battery
Microcontroller (**ESP32** or **Arduino**)
Motors
Purpose: To distribute the weight evenly and improve the robot's stability.

2. Leg Design

Number of legs: **4**
Each leg consists of:
Upper Leg
Lower Leg

3. Joints and Degrees of Freedom (DOF)

Each leg has:
Hip Joint = 1 Degree of Freedom (DOF)
Knee Joint = 1 Degree of Freedom (DOF)

Therefore:
2 DOF per leg
8 DOF for the entire robot

4. Motor Selection

Recommended Motor: Servo Motor

Reasons:
Provides precise angle control.
Suitable for walking movements.
Easy to program and control.

5. Preliminary Torque Calculation

Assume:
Leg mass = **0.5 kg**
Leg length = **0.2 m**
Gravity = **9.81 m/s^2**

Formula:
Torque = Force × Distance

First, calculate the force:
$F = m \times g = 0.5 \times 9.81 = 4.905$ N

Then calculate the torque:
$T = 4.905 \times 0.2 = 0.981$ N·m

Required torque ~ 1 N·m

6. Stability and Center of Gravity

The center of gravity should be located near the middle of the robot's body.
The battery should be placed at the bottom of the frame to reduce the risk of tipping over.
The legs should be spaced appropriately to improve stability.

7. Proposed Walking Gait

Walking Gait (Diagonal Gait)

Sequence:
Front Right Leg + Rear Left Leg
Front Left Leg + Rear Right Leg

This gait provides good stability while walking.

8. Expected Mechanical Challenges

Uneven weight distribution.
Insufficient motor torque.

High friction in the joints.
Frame bending under heavy loads.
Leg slippage on smooth surfaces.
High power consumption.

Figure ,Reference design of a quadruped robot.